

Invoice No.	97131
Bill of Lading	482637
Customer No.	8945
Customer P.O.	21004-02

NUCOR-YAMATO STEEL CO.

P.O. BOX 1228: BLYTHEVILLE, AR 72316

CERTIFIED MILL TEST REPORT

100% Melted and Manufactured in U.S.A

All Shapes produced by Nucor-Yamato Steel are cast and rolled to a fully killed and fine grain practice

Date 06/22/2021 6:55 AM

ASTM A992/A992M-20 A572/A572M GR50-18
ASTM A709/A709M-18 GR50 (345)
ASTM A709/A709M-18 GR50S (345S)
CSA G40.21-13 50WM (345WM)
ASTM A6/A6M-19

Item#	Item Description	QTY	Heat#	Mechanical Properties								Chemical Properties															
				Yield to Tensile Ratio	Yield Strength	Tensile Strength	ELONG	Charpy Impact				C	Mn	P	S	Si	Cu	Ni	Cr	Mo	V	Cb	CE	Sn	Pcm	Cl	
								Temp		Impact Energy	Loc																
					KSI	KSI	%	° F	ft•lbf																		
				MPa	MPa	%	° C	J																			
1	W12X065.0 67 ft 0 in W310X97 (20.42 m)	2	541081	0.80 0.80	60 61 414 418	75 75 518 519	26 25						.07	1.37	.014	.021	.28	.33	.18	.19	.07	.05	.003	.39	.02	.19	
2	W12X065.0 67 ft 0 in W310X97 (20.42 m)	2	541083	0.79 0.81	58 59 403 408	74 73 511 505	24 24						.06	1.26	.011	.030	.28	.34	.15	.11	.05	.04	.001	.34	.02	.16	
3	W14X090.0 48 ft 0 in W360X134 (14.63 m)	12	543330	0.76 0.76	52 52 357 358	68 68 467 472	27 26						.07	.91	.012	.024	.18	.33	.13	.08	.04	.00	.018	.27	.02	.14	
4	W14X090.0 48 ft 0 in W360X134 (14.63 m)	8	543333	0.76 0.75	51 51 351 350	67 67 464 464	26 27						.07	.92	.015	.027	.19	.35	.13	.10	.04	.00	.015	.28	.02	.15	

ELONGATION BASED ON 8.00 INCH GAUGE LENGTH

Pcm= C+Si/30+Mn/20+Cu/20+Ni/60+Cr/20+Mo/15+V/10+5B(B=Approx .0005)

Corrosion Index= 26.01(%Cu)+3.88(%Ni)+1.2(%Cr)+1.49(%Si)+17.28(%P)-7.29(%Cu)(%Ni)-9.10(%Ni)(%P)-33.39(%Cu)^2

ISO 9001:2015 certified (Registration #0985-07).

Meets mechanical lab independence requirements of EN10204 type 3.1.

The Charpy machine striker geometry used by Nucor-Yamato Steel is the 8 mm (0.315")striker (KV₈) per ASTM A370 Section 22.1.2 and ISO 148-1 Section 7.3.

CARBON EQUIVALENT CE= C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15

Mercury has not been used in the direct manufacturing of this material

Boron has not been intentionally added and will not exceed 0.0008%

This material was produced in accordance with the Nucor-Yamato Steel Quality Manual.

I hereby certify that the contents of this report are accurate and correct. All test results and operations performed by this material manufacturer are in compliance with the requirements of the material specifications, and when designated by the purchaser, meet the applicable specifications.

Dany Lennell

Chief Metallurgist

State of Arkansas
County of Mississippi
Sworn to and subscribed before me

on 2021-06-22
My commission expires on 07/17/2023

